

Skills Summary

Fitting and Interpreting Exponential Models

Use this reference while completing your worksheet or when studying.

Every model here is $y = a \cdot b^t$: a is the value at $t = 0$, b is the factor multiplied on each time step. Fitting finds a and b from data; interpreting reads them back as a starting amount and a rate.

Skill 1 — Fitting a and b from two points

Method: write one equation per data point, then *divide* the equations so a cancels: $\frac{ab^{t_2}}{ab^{t_1}} = \frac{y_2}{y_1}$.

Take the $(t_2 - t_1)$ -th root for b , then substitute back into either equation for a .

Example: Fit $y = a \cdot b^t$ through $(0, 50)$ and $(3, 86.4)$.

Step 1. Two unknowns, two points — divide the equations so the unknown a cancels and only b is left.

Step 2. $a = 50$ from the first point, so $50b^3 = 86.4$, giving $b^3 = 1.728$ and $b = 1.2$.

Try it: Fit $y = a \cdot b^t$ through $(0, 80)$ and $(3, 270)$: find b .

check: $80 \cdot 1.5^3 = 270$

Skill 2 — Interpreting a and b

Reading the model: a is the amount at $t = 0$ (in the units of the quantity), and b is the multiplier per time step.

- $b > 1$: growth, with rate $r = b - 1$ per step.
- $b < 1$: decay, with rate $r = 1 - b$ per step.

The rate is a percentage *of the current amount*, so the change per step is not a constant number of units.

Example: $A(t) = 600 \cdot (1.07)^t$. State the growth rate per step.

Step 1. The base is a multiplier, so compare it to $1 + r$ rather than reading it as an amount.

Step 2. $1.07 = 1 + r$ gives $r = 0.07$, a 7 percent increase per step, applied to whatever the current amount is.

Try it: $A(t) = 900 \cdot (0.92)^t$. State the decay rate per step.

check: $900 \cdot 0.8 = 720$

Skill 3 — Solving for time with logarithms

Method: isolate the power first, then take logs: $ab^t = y \Rightarrow b^t = \frac{y}{a} \Rightarrow t = \frac{\ln(y/a)}{\ln b}$.

For doubling, $y/a = 2$, so $t = \ln 2 / \ln b$ — independent of a .

Example: When does $600 \cdot (1.07)^t$ reach 900?

Step 1. The unknown sits in the exponent, so isolate the power and take logarithms of both sides.

Step 2. $(1.07)^t = 900/600$, so $t = \frac{\ln(900/600)}{\ln 1.07} = \mathbf{5.99}$ steps.

Try it: When does $500 \cdot (1.2)^t$ reach 1000?

check: $500 \cdot (1.2)^{5.99} \approx 1000$

Skill 4 — Rewriting the base

To the natural base: $b^t = e^{(\ln b)t}$, so $ab^t = ae^{kt}$ with $k = \ln b$. Here $k > 0$ means growth and $k < 0$ decay.

To a new time step: if one old step is n new steps, the new factor is $b^{1/n}$ (twelve months to a year: $b_{\text{month}} = b^{1/12}$).

Example: Write $(1.07)^t$ in the form e^{kt} .

Step 1. The two forms must agree for every t , so match the bases: e^k has to equal the given base.

Step 2. $e^k = 1.07$ means $k = \ln 1.07$, so $(1.07)^t = e^{t \ln 1.07}$ (about $e^{0.0677t}$).

Try it: Write $(0.92)^t$ in the form e^{kt} .

check: $e^{t \ln 0.92} \approx (0.92)^t$